

# transform\_data

## User Manual

Prepare Jira data for metrics and reports

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# 1 What is transform\_data?

transform\_data is the first module in the situation-report toolchain. It reads a raw data export from your issue tracker (Jira) and prepares the data for further analysis. The result is three **Excel files** that show how long issues spent in each workflow step and how the backlog evolved over time.

The program has a simple graphical user interface (GUI): no programming knowledge required. Select two files, click 'Run', and the Excel files are produced automatically.

## What transform\_data produces:

- **IssueTimes.xlsx**: All issues with milestone timestamps and minutes per workflow stage.
- **Transitions.xlsx**: Complete status history of all issues.
- **CFD.xlsx**: Daily entry counts per stage for the Cumulative Flow Diagram.

The generated Excel files are then used by the **build\_reports** module to create charts and reports.

## 2 Prerequisites and Installation

### 2.1 What needs to be installed?

transform\_data is delivered as a **portable package**. No separate Python installation is required.

- **Windows:** Python 3.11 is already included in the package — just unzip and run.
- **macOS / Linux:** On the first launch, a Python environment is set up automatically (approx. 1 minute, internet required). After that the app runs offline.

### 2.2 Starting the program

Double-click the appropriate launcher in the extracted folder:

- **Windows:** Double-click **TransformData.bat** — starts the GUI without a console window.
- **macOS:** Right-click **TransformData.command** → *Open* (once, to bypass Gatekeeper).
- **Linux:** In a terminal: `./TransformData.sh`

**Tip (Windows):** On the first launch, SmartScreen may show a warning. Click **More info** → **Run anyway**.

## 3 Input Files

transform\_data requires exactly two files: the Jira JSON export and the workflow definition file. Both are selected in the GUI via file dialogs.

### 3.1 Jira JSON Export

This is an export of all issues in a Jira project in JSON format. It contains the complete status history (changelog), metadata such as issue type and creation date, and the current status for each ticket.

How to generate the export from Jira is described in the **get\_data** module. The JSON file should not be edited manually.

**Typical filename:** ART\_A.json, MYPROJECT.json, etc.

**Size:** Depends on the number of issues; often several MB.

### 3.2 Workflow Definition File

The workflow file is a plain text file (.txt) that describes how the Jira statuses in your project map to logical process steps (stages). It also defines which stage marks the start of active work (First Date) and completion (Closed Date).

**Typical filename:** workflow\_ART\_A.txt, etc.

**Created by:** Your technical contact or Scrum Master -- once per project, rarely changed.

## 4 The Workflow Definition File

The workflow file controls how transform\_data interprets the Jira statuses in your project. You usually do not need to create this file yourself -- this chapter explains its structure so you can read and adjust it if needed.

### 4.1 Structure

Each line describes either a **stage** (a process step) or a **marker** (a milestone). Blank lines are ignored.

#### Example:

```
Funnel:New:Open:To Do
Analysis:In Analysis:Estimated
Implementation:In Implementation:In Progress
Done:Canceled
<First>Analysis
<InProgress>Implementation
<Closed>Done
```

| Format              | Meaning                                                                                |
|---------------------|----------------------------------------------------------------------------------------|
| Stage:Alias1:Alias2 | Stage with Jira status names mapped to it. The first name is the canonical stage name. |
| <First>Stage        | This stage marks the start of active work (First Date).                                |
| <InProgress>Stage   | This stage sets the Implementation Date.                                               |
| <Closed>Stage       | This stage marks completion (Closed Date).                                             |

### 4.2 Stage order

Stages are listed in the order they are traversed in the process. This order is used for the Cumulative Flow Diagram and for computing the Closed Date when stages are skipped.

### 4.3 Unmapped Jira statuses

If the Jira export contains statuses not defined in the workflow file, transform\_data issues a warning. Time in unmapped statuses is attributed to the last known stage (carry-forward).

## 5 The Graphical User Interface (GUI)

After starting, a window opens with file selection fields, options, and a log area for status messages.

### 5.1 Selecting files

Load the two input files:

- **JSON File** - Click 'Browse' and select your Jira export. The output folder and prefix are filled in automatically.
- **Workflow File** - Select the appropriate .txt workflow file for your project.

**Tip:** If you select the JSON file first, the output folder and prefix are pre-filled automatically.

### 5.2 Configuring output

| Field         | Meaning                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------|
| Output Folder | Directory where the three Excel files are saved. Default: directory of the JSON file.             |
| Prefix        | Name prefix for output files. 'ART_A' produces ART_A_IssueTimes.xlsx, ART_A_Transitions.xlsx, and |

### 5.3 Starting the transformation

Click '**Run**'. The program reads the data, computes all values, and saves the three Excel files. Progress and any warnings appear in the log area.

**Warnings in the log:** A warning about unmapped statuses does not mean the transformation failed -- the output files are still produced.

### 5.4 Language and Help

Use the **Options** menu to switch between German and English. All labels update immediately. Under **Help** → **Manual**, this manual opens in the active language in your browser.



## 6 Output Files

transform\_data produces three Excel files. These files serve as input for build\_reports and should not be edited manually.

### 6.1 IssueTimes.xlsx

The main output file. Contains one row per issue with all time data.

| Column              | Content                                           |
|---------------------|---------------------------------------------------|
| Project             | Project key (e.g. ART_A)                          |
| Key                 | Issue key (e.g. ART_A-123)                        |
| Issuetype           | Issue type (e.g. Feature, Bug, Story)             |
| Status              | Current Jira status                               |
| Created Date        | Issue creation date in Jira                       |
| First Date          | First entry into the <First> stage                |
| Implementation Date | First entry into the <InProgress> stage           |
| Closed Date         | Completion timestamp (empty = still open)         |
| Stage columns       | One column per stage: minutes spent in that stage |
| Resolution          | Closure type from Jira                            |

### 6.2 Transitions.xlsx

Contains the complete status history of all issues -- one entry per status change. Useful for detailed analysis of individual issues.

| Column     | Content                                              |
|------------|------------------------------------------------------|
| Key        | Issue key                                            |
| Transition | Stage name (or 'Created' for the creation timestamp) |
| Timestamp  | Date and time of the status change                   |

### 6.3 CFD.xlsx

Contains, for each calendar day, the number of issues that entered each stage on that day. build\_reports accumulates these values cumulatively to produce the Cumulative Flow Diagram.

## 7 How are Dates and Times Calculated?

This chapter explains the rules transform\_data uses to determine milestone dates (First Date, Closed Date) and stage times.

### 7.1 Stage times

| Situation                              | Rule                                                                              |
|----------------------------------------|-----------------------------------------------------------------------------------|
| Issue created but no status change yet | Time from creation to the first status change is attributed to the initial stage. |
| Status change to an unmapped stage     | Time continues in the last known stage (carry-forward). No time is lost.          |
| Current status                         | The last known stage accumulates time until processing time.                      |

### 7.2 First Date

The First Date is set when an issue first enters the stage marked **<First>** in the workflow file. It represents the start of active work.

**Skipped First stage:** If an issue enters a stage after the **<First>** stage without entering it directly, the entry timestamp of that later stage is used as the First Date.

### 7.3 Closed Date

The Closed Date is set when an issue enters the stage marked **<Closed>**. If an issue was closed and re-opened multiple times, the **last** closing timestamp is used.

**Re-opened issues:** If an issue is currently in a stage before **<Closed>**, no Closed Date is set -- even if the stage was reached previously.

**Never-worked issues:** Issues without a First Date (jumped directly from To Do to Closed) receive no Closed Date and are not counted in Flow Velocity.

## 8 Frequently Asked Questions

***Q: An issue has no First Date -- why?***

A: The issue never reached the stage or any stage after it. In build\_reports it is counted as 'To Do'.

***Q: An issue has no Closed Date -- why?***

A: Possible reasons: (1) The issue is still open. (2) It has no First Date. (3) It was re-opened after closing. (4) The stage was skipped with no later stage in the workflow.

***Q: The log shows a warning about unmapped statuses.***

A: Some Jira statuses in the export are not defined in the workflow file. Processing still completes. Contact your technical contact to update the workflow file.

***Q: The output files are not created.***

A: Check that the output folder exists and you have write permissions. Make sure the files are not open in Excel.

***Q: Which file do I use for build\_reports?***

A: IssueTimes.xlsx is required for all metrics except CFD. CFD.xlsx is additionally needed for the Cumulative Flow Diagram.

***Q: How often do I need to run transform\_data?***

A: Whenever you need updated data from Jira.

***Q: What does 'carry-forward' mean?***

A: Time spent in an unmapped status is attributed to the last known stage -- no time is lost.

## 9 Glossary

| Term                | Explanation                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------|
| Carry-forward       | Time in an unmapped status is attributed to the last known stage.                                  |
| CFD                 | Cumulative Flow Diagram -- shows backlog evolution over time by stage.                             |
| Closed Date         | The date an issue was completed.                                                                   |
| First Date          | The date active work began on an issue.                                                            |
| Implementation Date | The date development work started.                                                                 |
| Issue               | A ticket in the issue tracker (e.g. a Jira card).                                                  |
| JSON                | A simple text format for structured data.                                                          |
| Marker              | Lines in the workflow file that designate a stage as a milestone: <First>, <InProgress>, <Closed>. |
| Prefix              | Name prefix for output files (e.g. 'ART_A').                                                       |
| Stage               | A logical process step mapped to one or more Jira statuses.                                        |
| Workflow file       | Text file describing the stages and markers of a project.                                          |